CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

B. Tech. (EC) Semester - III Examination, 2010
EC – 203 Solid State Electronics

Date: 29-11-2010, Monday **Time: 10.30am to 1.30pm** **Maximum Marks: 70**

Instructions:
1. Attempt all the questions.
2. Figures to the right indicate *full* marks.
3. Make suitable assumptions and neat figures if required.

SECTION- I

Q1 Discuss the following terminologies in brief. **[10]**

(1) Pure Semiconductor
(2) Mean Free Path
(3) Break Point Voltage
(4) Pinch – off Voltage
(5) Thermistors

Q2 For an n-channel silicon FET with $a = 3 \times 10^{-4}$ cm and $N_D=10^{15}$ electrons / cm^3, find (i) the pinch off voltage and (ii) the channel half-width for $V_{GS} = (½)*V_p$ and $I_D = 0$. Relative dielectric constant = 12. Free space permittivity = 8.849×10^{-12} F/m. Assume and mention necessary values, if required. **[03]**

Q3 (a) Explain injected minority carrier charge and define electrical field due to injected minority carrier charge. **[08]**
(b) Explain transconductance (g_m) and drain resistance (r_d) for the FET small signal Model. **[06]**

OR

Q3 (a) Explain generation and recombination of charges. Also prove continuity equation. **[08]**
(b) Explain insulator, semiconductor and metals. **[06]**

Q4 Attempt any two. **[08]**
(a) Draw neat sketch of two-stage RC coupled amplifier. Explain the need of using capacitor in RC coupled amplifier.
(b) Write Short Note: Current components in PN junction diode.
(c) Explain Capacitive filters.

SECTION - II

Q5 Attempt any four questions. **[08]**

(1) Mention significance of Biasing the transistor
(2) Enlist advantage of BJT over vacuum tube.
(3) Derive relationship between β and α from known relationship between I_E, I_C and I_B.
(4) With respect to transistor explain rise time and fall time.
(5) Justify: Voltage divider bias is utilized widely compare to fixed bias technique for biasing the transistor.

Q6 (a) Why the name *h-parameters* is been given for the parameters in hybrid model of transistor amplifier? For single stage CE amplifier, derive equations for Current Gain (A_i) and Input Impedance (Z_i) for equivalent h – parameter model. **[07]**

(b) For the circuit shown in *figure 1*, with I_C = 2 mA and β = 100, calculate R_E and V_{CE}. Assume and mention necessary values, if required. **[08]**

OR

Q6 (a) Using sketches for voltage divider biased CE amplifier, mention guidelines for converting CE transistor amplifier to its h-parameter model. **[07]**

(b) For the circuit shown in *figure 2*, with β = 100 for the silicon transistor, calculate V_{CE} and I_C. Assume and mention necessary values, if required. **[08]**

Q7 Attempt any two. **[12]**
(a) Explain the Miller theorem and its proof.
(b) With neat sketch explain output characteristics of Common Base Amplifier.
(c) Justify the need of bias compensation. Explain any diode compensation technique with necessary details.
(d) Perform dc analysis of fixed bias technique for biasing of transistor.

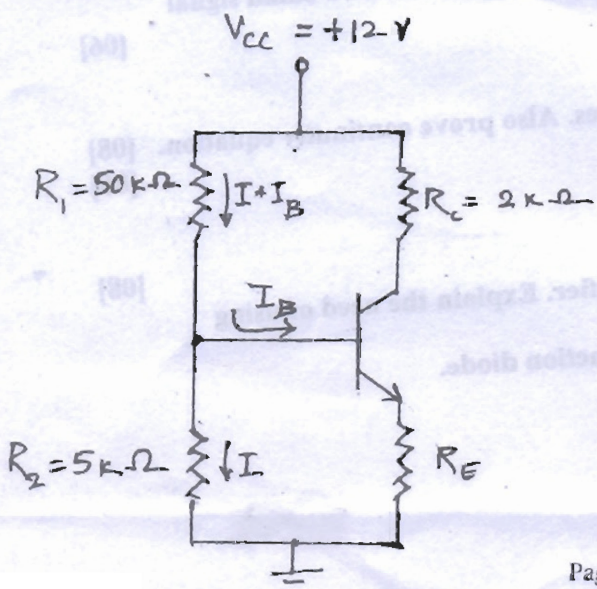

FIGURE-1

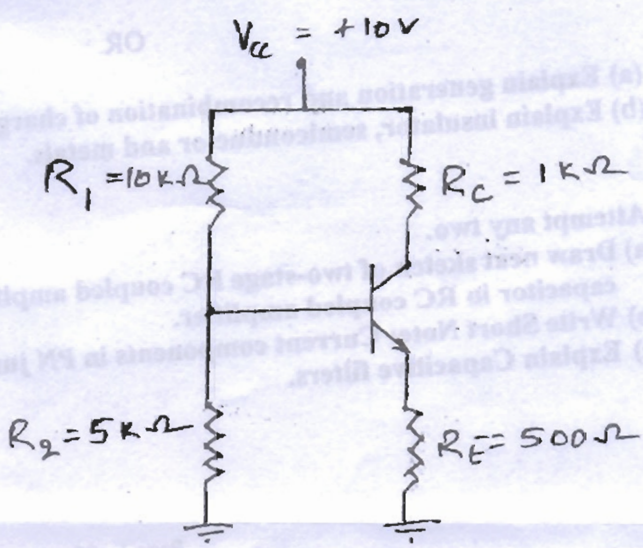

FIGURE-2